Part III Project Notebook

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Contents

1 Introduction

This is my project log, which is updated regularly to document my progress, thoughts, and reflections throughout the duration of my project. It serves as a personal journal where I can track my journey, challenges faced, solutions implemented, and insights gained.

It also pulls in my progress and goals from the shared Google Doc that I have with my supervisor, acting as a bridge between my personal reflections and the collaborative aspects of my project.

2 Plan for the project in outline

- Implementing the traditional method(s) for estimating sea level change from satellite altimetry observations, and investigating its accuracy. For global mean sea level, this method is just to spatially average sea surface height changes over the oceans.
  - Roughly first 2-4 weeks

- Application of Bayesian inversion methods that incorporate sea level physics. First done for estimates at a single time. Comparing new methods against old.
  - Roughly second half of Michaelmas term
- Extension to consider time-dependent estimates.
  - Running over Christmas break
  - Consider building in feed forwards mechanisms (Kalman filter-like)
- Possible extension to consider:
  - Adding ice altimetry data and/or other data types
  - Effect of knocking out %'s of the networks (e.g. esp tidal gauges → look at their average downtime)
  - Comparison of the data types (more data vs more types of data)
  - Consider if different data types have different abilities to resolve features (e.g. tidal gauges trade off within East and West Antarctic Ice ... does this exist within altimetry data?)
  - Early Lent

3 Weekly progress/goals

Mainly here to keep track of what was spoken about so I can (a) look through paper/written notes and (b) understand changes to code based on meetings

3.1 Meeting 1: 13/Oct

- Start reading through the literature esp. focusing on:
  - Horton 2018 for general overview
  - Lickley 2018 for sea level to surface
  - The Reciprocity Paper (TRP) §4.6 (and then the references within this)
- Look at implementing equivalent of eq. 81 from TRP to convert the SL change from fingerprint to SSH to then average this over the ocean for GMSL change

3.2 Meeting 2: 17/Oct

- Discussion of scientific method
- Explore the error space for conventional methods
  - Error for each ice sheet over satellite availability bands
  - Ternary space for major ice sheet contributions at satellite availability bands
- Future explore adding:
  - Noise based on satellite sampling
  - Ocean dynamic signals
- Writing useful scripts/functions that can be reused in the future

3.3 Meeting 3: 21/Oct

- Went through error exploration code
- Look into how changing lengths scales can change end distributions
- Explore forcibly pushing forward distribution

3.4 Meeting 4: 24/ Oct

- Work through PyGeoInf tutorials 1→5
- Discussed distribution mapping and pushing forward through function theory
- Parallelise the samples function
- Look at non-zero means

3.5 Meeting 5: 31/Oct

- Spoke through random field operators and pushing forward distributions
- Spoke through workflow of error quantification
- Speak to Dan wrt. Ocean dynamics and ice sheet change sensible distributions

    - Ratios of contribution to error of melt vs dynamics vs noise (… another ternary?)
- Goal to work on implementing a workflow for estimating errors

3.6 Meeting 6: 3/Nov

- Discussed briefly:
- Extracting statistics from gaussian_measure.GaussianMeasure class
- Linear transform for Measure: affine_mapping(*, operator: LinearOperator = None, translation: Vector = None) → GaussianMeasure
- Direct sum
    - Of R^n space
    - Of gaussian measures
- Work on extracting out script into functions and adaptable workflow

3.7 Meeting 7: 7/Nov

4 Literature log

5 Thoughts log

5.1 2025-10-10 22:01

To what extent can the noise/artifacts that were filtered out by @coulson2022 be build into the output to see what the "real world" observable would be?

5.2 2025-10-17

For parallel processing, using joblib's Parallel and delayed is much simpler than multiprocessing.Pool. Just wrap the function call in delayed() and pass to Parallel.

Tried using ternary-plot but found mpternary worked much better.

5.3 2025-10-18

Build a load across multiple latitudes and then plot this against error at different satellite segments to see how error varies with load and latitude.

5.4 2025-10-26

Using ternary plots works well but it is important to provide contextual side plots that show a selection from the ternary location in actual space.

For example for melt contribution plots show a side plot of what each ternary space plot looks like on a map.

Similarly having bands of satalite latitude ranges showing what is being sampled by different plots can help contextualise the differences seen in distribution ternary plots.

6 Code notes

6.1 Workflows

6.1.1 Deterministic space

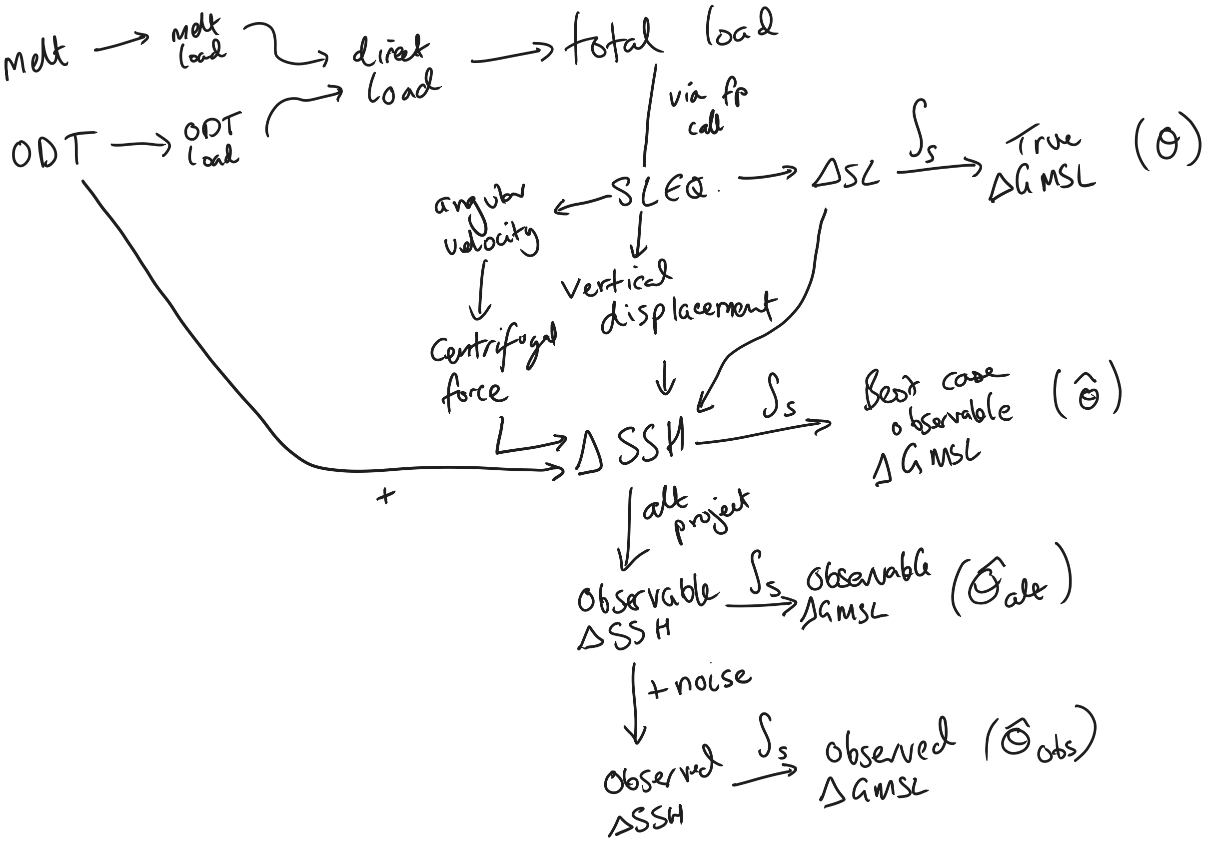

Figure 1: alt text

6.1.2 Measure space

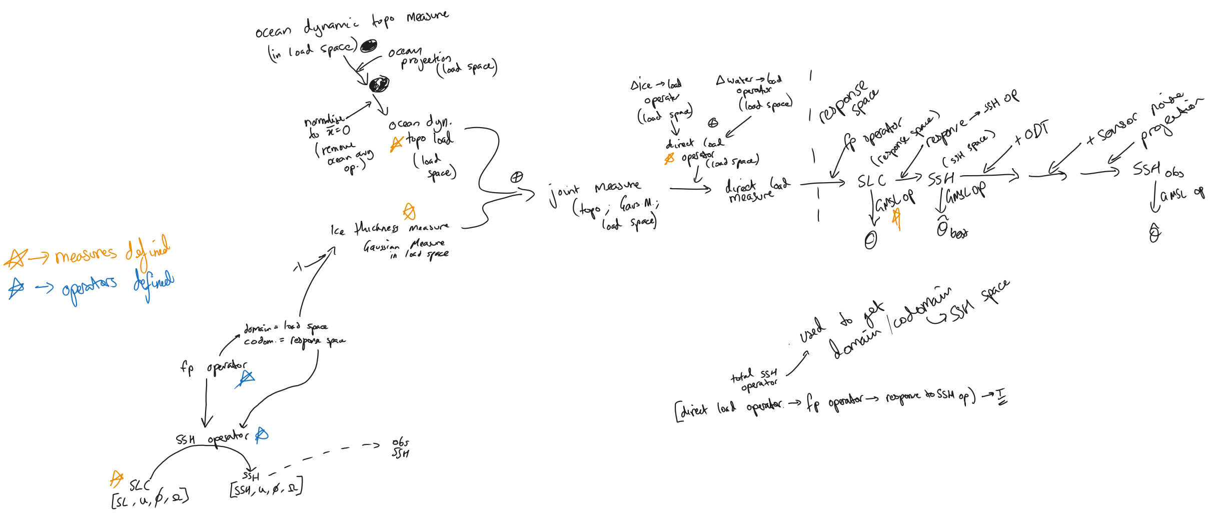

Figure 2: alt text

7 Maths

7.1 Maths of SSHC and SLC

$$\Delta SL = \Delta SSH - \Delta R$$

Change in topo

$$= u + \frac{\psi}{g}$$

displacement

centrifugal potential

ocean func $= \begin{cases} \text{is ocean} = 1 \\ \text{else} = 0 \end{cases}$

f(direct load)

total load = direct load $+ \rho_w C \Delta SL$

$\Delta GMS\_$ weighting:

e.g. $\Delta GMSS = \int_S w \cdot \Delta SSH \, dS$

where w defined:

$$1 = \int_S w \, dS$$

Figure 3: Maths of SSHC and SLC

7.2 Probability Distributions

Working through of probability distributions

$X \to$ sample space

$x \in X$ $\quad x$ is a point in X

$\underbrace{\mathcal{N}(\bar{x}, c)}$ where $\bar{x} \in X$

Gaussian distribution w/ expectation $\bar{x}$ and covariance c

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Let $A : X \to Y$ and is linear

matrix (which can act like a function)

Samples from X: $\{x_1, x_2, \dots, x_n\}$

map to: $\{Ax_1, Ax_2, \dots, Ax_n\}$ in Y space

Y drawn from $\mathcal{N}(A\bar{x}, ACA^T)$

$$y = f(x) = a + Ax$$ $\leftarrow$ NB affine function, non linear

$$x \sim \mathcal{N}(\bar{x}, c)$$

$$y = a + Ax \sim \mathcal{N}(a + A\bar{x}, AcA^T)$$

in vector / Matrix notation:

$A \to$ matrix

$a \to$ vector translation

Covariance matrix: $\underline{\underline{C}}$

σ^2: variance $\leftarrow$ in square units

$$\underline{\underline{C}} = \sigma^2 \underline{\underline{I}}$$ $\leftarrow$ in $\mathbb{R}^1 \to \underline{\underline{C}} = \sigma^2 \underline{\underline{I}} \to \underline{\underline{C}} = \sigma^2 (1)$

$$\downarrow$$

$$\underline{\underline{C}} = \sigma^2$$

is $\mathbb{R}^n$ $\to$ $\underline{x} \sim \mathcal{N}(\underline{\bar{x}}, \underline{\underline{C}})$

Key variables:

$$E[\underline{x}] = \underline{\bar{x}} = \int \underline{x}\, p(x)\, dx$$

$$= \frac{1}{N} \sum_{i=1}^{N} x_i$$

$$\mathrm{Var}[x] = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

$$\mathrm{Cov}[x_i, x_j] = \frac{1}{N} \sum_{k=1}^{N} (x_i^k - \bar{x}_i)(x_j^k - \bar{x}_j)$$

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \end{pmatrix} \quad \begin{matrix} x_1 \sim \mathcal{N}(\dots) \\ x_2 \sim \mathcal{N}(\dots) \end{matrix}$$

Figure 4: Distributions

Working through pushing functions

8 Extra Maths

This is maths that learnt along side the project that was relevant.

$\underline{x} \rightarrow$ vector
$\underline{\underline{A}} \rightarrow$ matrix

Transpose:

along trace

Let A be matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$

$A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$

$3\times 2 \quad \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$

$\dim A = m \times n$
domain $A = m$
Codomain $A =$ domain $A^T = n$

$D(X) \rightarrow$ domain of space X (83)
$\dim X \rightarrow$ dimensions of space X (54)
$T^* \rightarrow$ Hilbert-adjoint operator of T (196)
$T^\times \rightarrow$ Adjoint operator of T (232)
$X^* \rightarrow$ Algebraic dual space of vector space X (106)
$\langle x, y \rangle \rightarrow$ inner product of x and y (128)

Direct Sums:

$$\underline{\underline{A}} \oplus \underline{\underline{B}} = \begin{bmatrix} \underline{\underline{A}} & 0 \\ 0 & \underline{\underline{B}} \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & b_{m1} & \cdots & b_{mn} \end{bmatrix}$$

for vector space $\underline{a} \oplus \underline{b} \rightarrow \mathbb{R}^a \oplus \mathbb{R}^b = \mathbb{R}^{a+b}$

$\underline{a} \oplus \underline{b} \rightarrow \begin{pmatrix} \underline{a} \\ \underline{b} \end{pmatrix}$

if $\underline{a} \in A$ and $\underline{b} \in B \rightarrow \begin{pmatrix} \underline{a} \\ \underline{b} \end{pmatrix} \in X \oplus Y$

Of distributions:

$\mu \sim N_X(\bar{u}, \underline{\underline{Q}})$
$\gamma \sim N_Y(\bar{v}, \underline{\underline{R}})$
$\mu \oplus \gamma \sim N_{X\oplus Y}(\bar{u} \oplus \bar{v}, \underbrace{\underline{\underline{Q}} \oplus \underline{\underline{R}}}_{\begin{pmatrix} \underline{\underline{Q}} & 0 \\ 0 & \underline{\underline{R}} \end{pmatrix}})$

Figure 5: Matrix notation after (TEXTBOOK)

Vector space:

X

$x \in X$

Must have two operators:

vector addition: $x + y \in X$
Scalar multiplication: $ax \in X$, $a \in \mathbb{R}$

e.g. $\mathbb{R}^n$: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

addition: $\underline{x} + y \rightarrow \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$

e.g. continuous functions on $[0,1]$:

$$(u + v)(x) = u(x) + v(x)$$
$$(au)(x) = au(x)$$

Basis for a vector

- → what underlies all vectors → c.f. diffent grid systems in QGIS
- → spans the vector space
- → every vector in a space can be written as linear combination of the basis vectors
- → linearly independent

eg $\mathbb{R}^n$: $\underline{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, $\underline{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$, $\underline{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$

$\forall \underline{x} \in X$

$$\underline{x} = \sum_{i=1}^{n} x_i \underline{e}_i$$

Figure 6: Vector spaces 1